4 **(a)** Describe one condition necessary for observable two-source interference fringes to be formed.

.....

..... [1]

(b) Two microwaves transmitters produce waves of the same frequency are placed at P and Q which are at a distance of 2*d* apart. Points Y and Z are equidistant from O. The line YXZ is perpendicular to OX, as shown in Fig. 4.1.

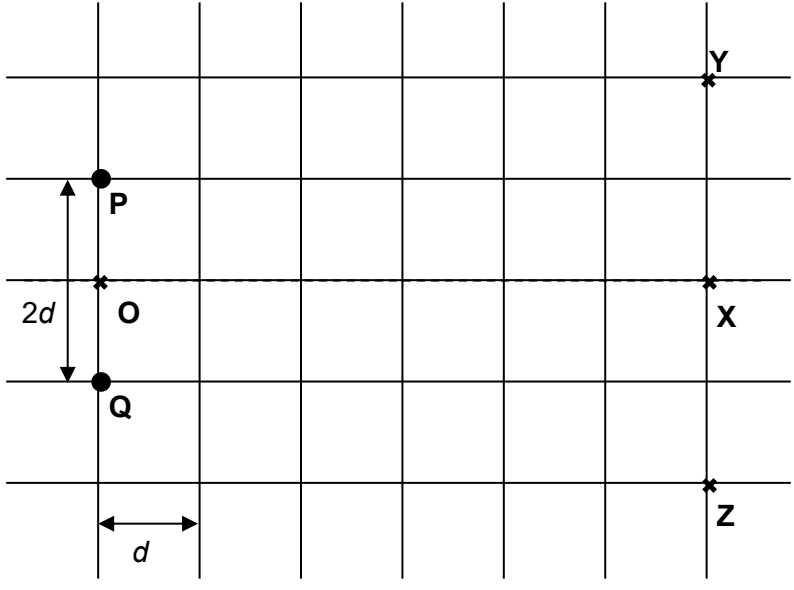

Fig. 4.1

[Turn over

(i) The waveforms of the microwaves from P and Q arriving at point Y vary with time as shown in Fig. 4.2.

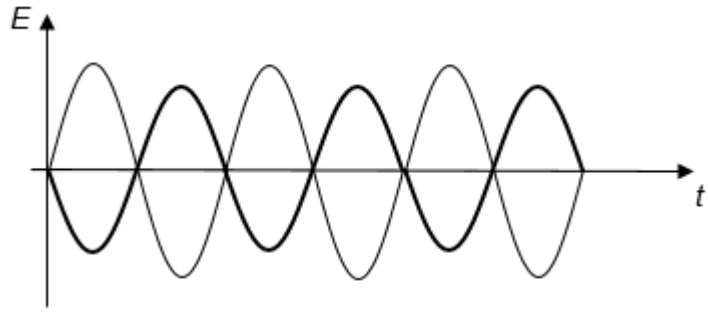

Fig. 4.2

1. State and explain if the waves arriving at Y are coherent.

.....

.....

..... [2]

2. Explain why a minimum intensity is detected at Y.

.....

..... [1]

(ii) Show that the path difference of the waves arriving at point Y from P and Q is 0.625*d*.

[1]

(iii) As a detector is moved along a straight line from X to Y, it encountered three intensity maxima, including the maximum at X.

Determine the frequency of the wave in terms of *d*.

frequency = ...Hz [3]